

PHALABORWA, South Africa

WMO: 681900

Lat: 23.937S

Lon: 31.155E

Elev: 437

StdP: 96.19

Time Zone: 2.00 (E02)

Period: 98-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	8.0	-6.0	2.4	21.8	-2.9	3.1	23.1	7.3	18.0	6.5	18.2	0.6	320	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.9	36.8	21.3	35.4	21.1	34.1	20.9	26.2	30.5	25.3	29.7	24.6	29.0	2.5	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.3	28.3	24.1	20.1	27.5	23.2	19.0	26.7	84.2	30.6	80.3	29.5	76.8	29.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.3	5.4	DB	4.3	39.8	1.5	1.3	3.2	40.8	2.4	41.5	1.5	42.3	0.4	43.2	(4)
(5)			WB	2.5	26.0	1.4	1.6	1.5	27.2	0.7	28.2	-0.1	29.1	-1.1	30.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.3	25.9	25.9	25.1	22.4	19.9	17.7	17.0	19.0	21.6	23.2	24.6	25.3	(6)	
	DBStd	4.06	2.29	1.97	2.42	2.42	2.59	2.26	2.04	2.59	3.13	3.16	3.21	2.92	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	134	0	0	0	1	12	38	51	23	6	2	1	0	(9)	
	CDD10.0	4481	493	445	468	371	307	231	218	278	349	409	437	476	(10)	
	CDD18.3	1573	235	212	210	122	59	19	10	42	105	153	188	217	(11)	
	CDH23.3	15245	2177	1850	1785	938	633	308	243	602	1218	1554	1841	2096	(12)	
(13)	CDH26.7	6482	923	787	766	332	197	70	52	205	565	734	886	966	(13)	
(14)	Wind	WSAvg	2.1	2.3	2.3	2.0	1.8	1.4	1.5	1.7	2.0	2.5	2.5	2.3	2.3	(14)
(15) Precipitation	PrecAvg	442	81	83	58	18	6	6	5	3	5	26	63	93	(15)	
	PrecMax	1013	193	453	229	66	31	29	35	11	23	57	162	193	(16)	
	PrecMin	166	3	3	2	0	0	0	0	0	0	1	3	15	(17)	
	PrecStd	203	52	107	57	16	8	8	8	3	7	18	41	48	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	36.9	37.0	34.4	33.9	31.1	30.3	33.6	37.0	38.3	38.2	37.1	(19)	
		MCWB	24.2	22.8	22.5	20.7	18.1	15.3	14.2	16.2	17.3	20.0	21.2	22.3	(20)	
	2%	DB	34.3	34.5	34.7	32.4	30.9	28.9	28.1	31.1	34.6	35.7	36.1	35.5	(21)	
		MCWB	23.2	22.3	22.4	20.2	17.5	15.1	14.7	16.0	17.5	19.8	20.7	22.2	(22)	
	5%	DB	32.9	32.8	32.7	30.6	28.8	26.9	26.2	29.1	32.2	33.4	33.9	34.0	(23)	
		MCWB	22.9	22.0	21.6	20.1	16.9	15.4	15.0	15.3	17.3	19.5	20.8	22.2	(24)	
	10%	DB	31.2	31.3	30.9	28.5	26.9	25.0	24.1	26.9	30.1	31.0	31.9	32.0	(25)	
		MCWB	22.9	22.1	21.6	20.1	17.0	15.5	14.8	15.7	17.8	19.5	21.1	22.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.1	26.8	25.4	22.7	20.0	19.2	20.1	22.5	24.2	25.6	25.8	(27)	
		MCDB	30.9	30.4	31.4	31.5	29.1	24.5	23.8	25.5	30.7	31.5	31.0	31.8	(28)	
	2%	WB	26.0	25.9	25.6	23.6	20.7	18.7	18.1	18.8	20.9	22.3	24.1	24.6	(29)	
		MCDB	29.6	29.1	30.5	28.5	26.4	23.8	23.6	24.7	28.4	29.3	29.2	30.5	(30)	
	5%	WB	25.0	25.0	24.6	22.3	19.5	17.7	16.9	17.8	19.6	21.3	23.1	23.9	(31)	
		MCDB	29.0	28.7	28.6	26.2	25.2	23.0	22.0	24.2	27.3	29.0	28.8	29.7	(32)	
	10%	WB	24.1	24.1	23.5	21.2	18.4	16.7	15.9	16.9	18.7	20.4	22.3	23.1	(33)	
		MCDB	28.6	28.1	27.4	25.6	23.7	22.1	21.2	23.5	26.2	27.8	28.2	29.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	9.8	10.3	11.0	14.0	14.9	14.4	14.3	13.8	11.9	11.0	10.6	(35)	
		MCDBR	12.5	12.9	13.7	15.2	17.5	18.1	18.6	18.9	18.9	17.4	15.4	14.1	(36)	
	5% WB	MCWBR	3.5	3.6	4.0	5.7	6.3	7.5	7.7	7.3	6.5	5.4	4.6	4.1	(37)	
		MCWBR	10.1	9.1	10.2	11.7	13.3	14.3	14.7	15.2	14.8	13.4	11.7	11.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.410	0.402	0.396	0.381	0.351	0.335	0.337	0.391	0.453	0.463	0.429	0.423	(40)	
		taud	2.376	2.401	2.408	2.406	2.439	2.459	2.444	2.283	2.139	2.166	2.297	2.335	(41)	
	Ebn at Noon		935	929	905	874	861	856	867	850	838	863	912	925	(42)	
		Edh at Noon	131	125	119	109	97	91	95	122	153	156	141	137	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.12	5.98	5.34	4.58	4.22	3.77	3.94	4.63	5.23	5.48	5.80	6.02	(44)	
	RadStd		0.43	0.49	0.40	0.29	0.19	0.24	0.15	0.27	0.36	0.34	0.51	0.56	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page